# Aquae Sulis Music Venue:
## *Geotechnical Design Technical Report*

**Client:** University of Bath, Department of Architecture & Civil Engineering

Clayey Silt Geotechnical Solutions LTD

Group 3

Lewis Acott – Tom Grant – Dan Heard – Cadi Hicks Jones

## *Note of Thanks:*

We extend our gratitude to "WH Concrete Consultants" for their professional courtesy in providing us with the required loads for foundation design.

## *Table of Contents:*

# *Executive Summary*

Clayey Silt Geotechnical Solutions Ltd was employed by the University of Bath's Department of Architecture & Civil Engineering to carry out the geotechnical design work for the new 'Aquae Sulis' music venue brownfield development.

The client also employed a geotechnical contractor to acquire adequate information about the ground conditions present at the site, this information was then passed to us to carry out the geotechnical design.

After receiving the CPT, Triaxial & Oedometer data we analysed each component so that they could be utilised for design. The CPT data was converted into soil profiles so we could plot transects across the site.

We decided to use a single concrete driven pile for the minimum load (226kN). This had a design depth of **5000mm**, square cross section of **500x500mm** & a design capacity of **269kN** and would be predicted to settle by **19.69mm** over the design life of the structure.

For the maximum load (1.5Mn) we decide to use a pile group consisting of a triangular concrete 'raft' connected to **three 4000mm** deep, **400mm** diameter driven concrete piles. This foundation system provides a resistance of **1803kN** and is predicted to settle by **15.04mm** over the design life of the structure.

The client required no settlement greater than 30mm which has been met (**Max: 19.69mm**) and differential settlement of less than 10mm which has been met (**4.65mm**).

## *Kind Regards*

***LEWIS ACOTT, Senior Design Engineer, Clayey Silt Geotechnical Solutions LTD***

# *Site Investigation & Construction of Ground Profile*

*Creation of Soil Profiles:*

Clayey Silt Geotechnical Solutions Ltd was provided with sporadic CPT data from the Client via an external contractor. It is regrettable that the CPT test was carried out in such a manner that did not lean itself to forming soil transects. The CPT data was collected from seemingly random locations across the site as opposed to in a uniform gird, presumable to save on site investigation costs.

After GS Ltd processed the CPT data the company had to approximate the locations for two perpendicularly bisecting soil transects. The makeup of a soil profile can change significantly over small horizontal displacements and as a result GS Ltd took the decision to conservatively estimate the resistances present from each stratum to partially compensate for the lack of certainty in the makeup of the soil.

We appreciate that conservative assumptions lead to an element of overengineering however because of the nature of the CPT data we believe this to be a necessary precaution.

As a result, we selected a factor of safety of **2.5** for our foundation design.

*Processing the CPT Data:*

We received the spreadsheets of data for each of the CPT locations and divided them equally between our team of engineers. We were also provided the depth of the water table, so we first calculated and plotted the pore pressure throughout the depth of the CPT borehole.

We initially assumed a uniform bulk density throughout the soil profiles of **19.5kN/m$^3$** as this is a middling value within the overall range of bulk densities of soil types likely to be present. We believed that selecting a middling value was the most efficient way to start the iterative process of determining the true bulk densities of the soil because of its equal probability of proximity to the true values.

After selecting our initial bulk densities, we used Terzaghi's Effective Stress equation (Eq1) & total stress equations (Eq2) to calculate the effective stresses at **0.02m** intervals down the borehole.

$$\sigma' = \sigma - u \qquad \text{(Eq1)}$$

$$\sigma = \gamma_B h \qquad \text{(Eq2)}$$

At the same time using Excel we plotted the sleeve friction and the cone resistance, Figures 1 and 2 illustrating this data for CPT000000016347.

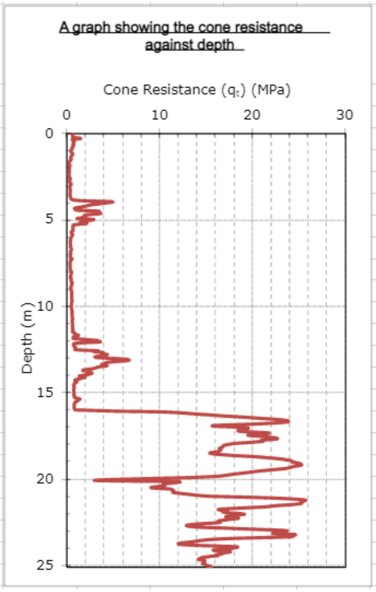



*Figure 1 – $q_t$ vs depth*

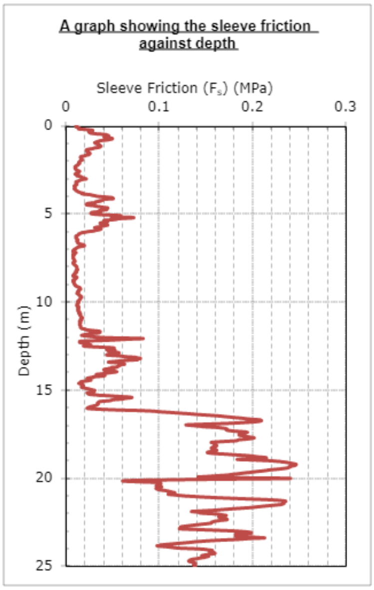



*Figure 2 –$F_s$ vs depth*

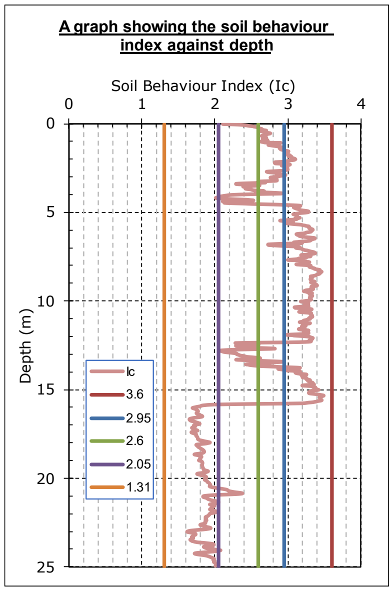



*Figure 3 – Soil Behaviour Index Chart*

To correspond with our bulk density, we initially assumed that n**=0.5.** Using this assumption and the data we had collected and calculated we were able to calculate the friction ratio, *Fr* and normalised tip resistance, $Q_{tn}$. *(Eqns 3 - 5)*

$$F_r = \frac{f_s}{(q_t - \sigma_{v0})} \qquad \text{(Eq3)}$$

$$Q_t = \frac{(q_t - \sigma_{vo})}{\sigma'_{v0}} \qquad \text{(Eq4)}$$

$$Q_{tn} = \left(\frac{q_t - \sigma_{vo}}{P_{atm}}\right)\left(\frac{P_{atm}}{\sigma'_{vo}}\right)^n \qquad \text{(Eq5)}$$

After calculating the variables, we then proceeded to plot the friction ratio against the normalised tip resistance on Robinson's Chart as shown in *Figure 4.*

Robinson's chart allows for a graphical interpretation of the makeup of the soil sample based on the assumptions however in this format it is not clear which points are corresponding to which depths. To combat this, we re-plotted the data against depth over our soil profile as shown in *Figure 4*. (see next page)

Figure 4 clearly shows the differences in soil types in each layer and was the critical component in each following iterative step. It allows us to select appropriate bulk densities for individual layers instead of assuming homogeneity throughout. The process described above can be repeated to plot a new Soil Classification Chart, leading to a more accurate chart to represent the layers present.

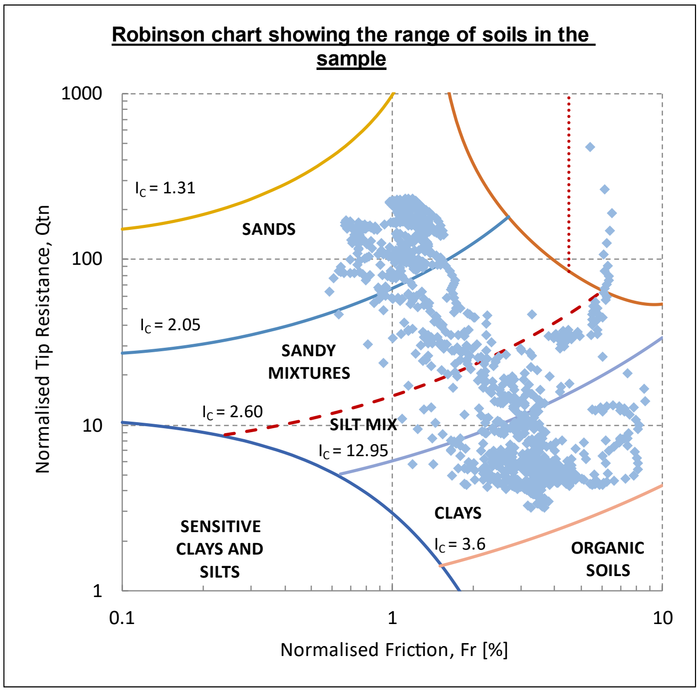



*Figure 4 - Robinsons Chart Depicting Soil Type*

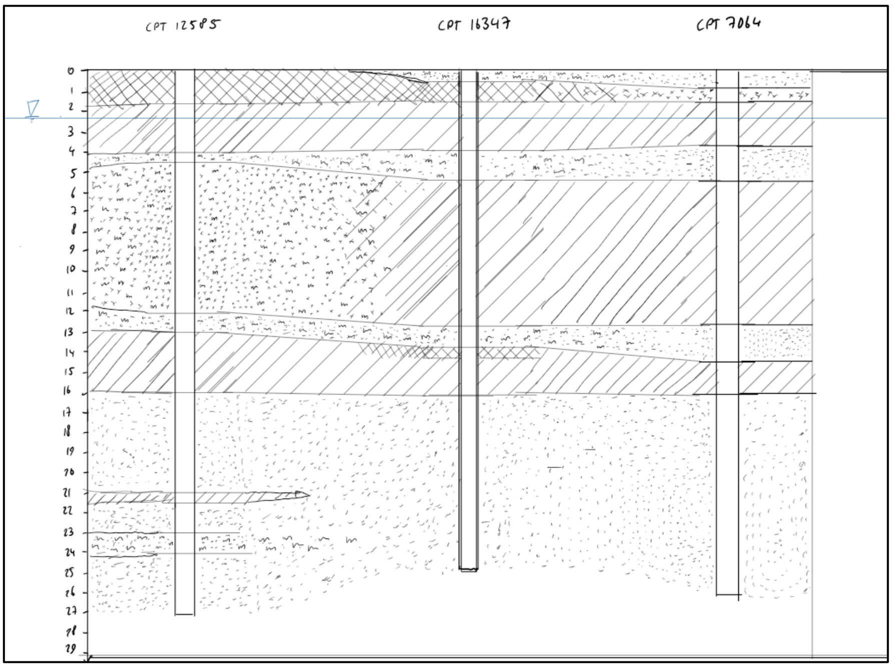
We repeated this iteration until the results converged in each layer giving us the 'real' soil properties present that we were to utilise for our foundation design as well as the depths at which they occur.

We used the converged data to plot soil profiles for each of the six CPT boreholes and arranged them into two perpendicular transects, one running broadly east to west and the other north to south. We then interpolated between adjacent CPT boreholes on the transects to determine the 'predicted' soil makeup between known data points.

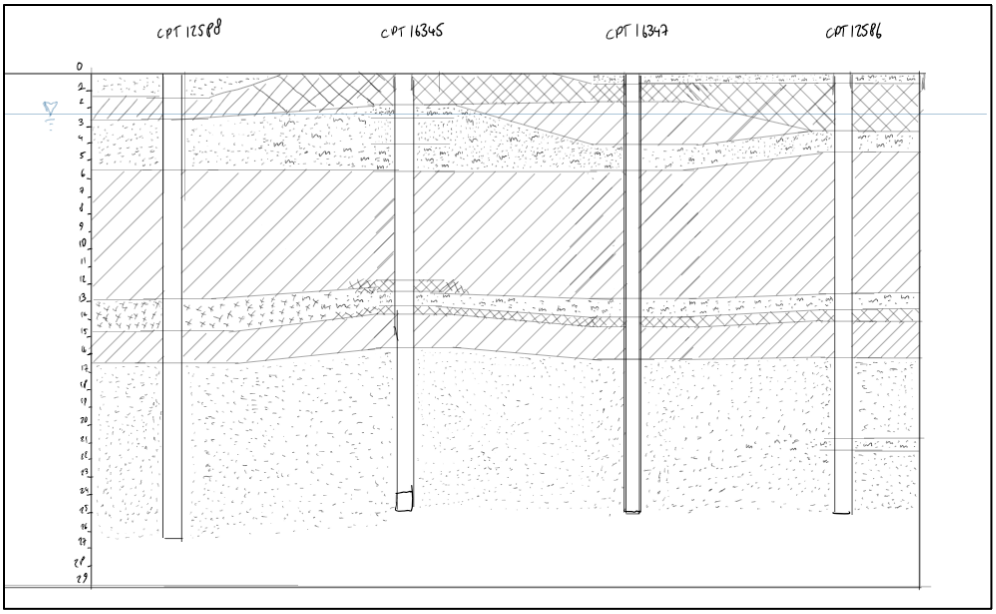
We then plotted the two transects to show a cross section through the site along both planes assuming a flat ground surface throughout. See the next page for both the North-South & East West soil profile transects.

## *Transect Key:*

## *East-West Transect:*




## *North-South Transect:*




## *CPT Selection for Foundation Design:*

Overlaying the soil profiles over the floorplan of the concrete structural system allowed us to identify the two CPTs most relevant to minimum & maximum loads:

Maximum Load Foundation: CPT00000007064

Minimum Load Foundation: CPT000000012585

# *Triaxial Data Analysis*

The Triaxial Test was carried out in accordance with BS 1377-1 for Sample preparation and BS 1377-8 for effective stress. Throughout the test, Force applied to the sample as well as displacement was recorded to visualise the stresses at failure e.g.  the failure plane.

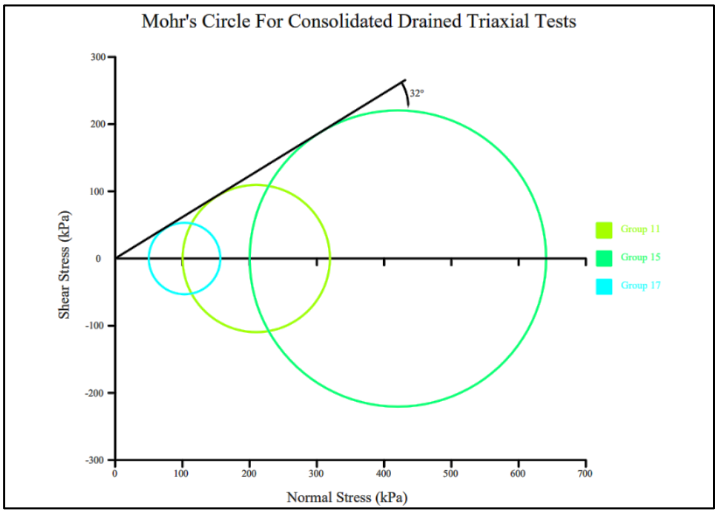



*Figure 3 - Mohr's Circle for Consolidated Drained Triaxial Tests*

The test undertaken was the Consolidated Drained test. All calculations were performed under the guidance of **BS EN ISO 17892-8:2018** to find the appropriate failure stresses. By finding the displacement of the soil sample as the deviator stress increased, it was possible to estimate the internal angle of friction by finding the failure envelope of each sample, and therefore the internal angle of friction. Each test was carried out using different pore pressures and cell pressures to gain a variety of Mohr's Circles (plotted below), allowing an average angle, $\phi'$ to be found. Each group's data was plotted to find an internal angle of friction of **32**°. Tests 11, 15 and 17 are shown here to best represent our findings.

# *Oedometer Data*

The Oedometer given was not used in the calculations for pile capacity or settlement. It was decided not to use this data, in favour of the CPT data as the primary design focus. The results given were used to calculate the Preconsolidation Pressure using Casagrande's method, as well as the coefficient of consolidation, $C_v$ using the Log-time method.

Since our piles were designed primarily to sit in sandy layers of soil, it seemed unwise to use data from such a limited sample, as it is unrepresentative of the strata.

In the case of a future redesign of the pile foundations, the Preconsolidation Pressure was found to be **625kPa.**

# Maximum Load: Foundation Design

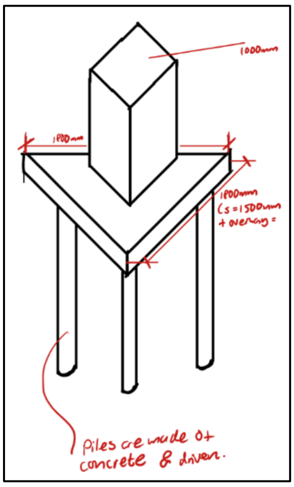



*Figure 5 - Illustration of the Maximum Load Pile Group*

***Background & Design Justification***

The highest load within the concrete structure which needs to be resisted by the soil is ≈**1.5MN**

Using our engineering judgement, we decided that it would be inappropriate to design a shallow foundation to resist this loading as it would require significant volumes of concrete, it would not be an efficient use of material and would constitute a significant amount of embodied carbon.

A single pile foundation was then considered however this would have to be very deep to generate sufficient shaft resistance and have a large diameter to generate a high enough base resistance. This option was made less suitable for our design due to the presence of a large clay layer beneath the desired location of this pile, clay is not conducive for generating large bearing capacities in piles and further increased the required depth.

The selected solution is a pile group. This is comprised of three shorter piles joined by a concrete raft that equally distributes the loading from the concrete superstructure. This solution avoids the need for very deep piles and takes advantage of a sand layer ~5m below the surface that provides significant bearing capacity.

The selected option reduces the capital cost as less material is needed, speeds up construction as the depth of the piles is significantly reduced and contributes to lower carbon emissions from the overall project because of reduced timescales and less material usage.

***Capacity Calculation Methodology***

After acquiring the load from WH Concrete Consultants, the next stage was calculating the size of the concrete raft at the base of the column. The purpose of this raft is to evenly distribute the load from the arch to every pile in the group. The raft had to be designed and set out in a manner that prevents against differential settlement with the pile group. This should be avoided because if the differential settlement is too great it can cause damage to the superstructure as it can cause deformations to the elements. These deformations have the capability to change or introduce eccentricities into load paths and contribute towards second order moment effects. Differential settlement also has potentially severe implications for serviceability, if the arch supported by the raft settles differentially it could impact the services and lighting systems above the music venue which are supported from the structure. In severe cases differential settlement can render a structure unusable.

Classification of pile groups is governed by *Equation 6* which relates the spacing of the piles acting together to their respective diameters. A combination of this equation and symmetrical spatial geometry were used to design the raft based on a group of three piles working together.

$$\frac{s}{d} \equiv 2 - 4 \qquad \text{(Eqn 6)}$$

Selecting an initial diameter of 400mm for each pile and a spacing (Centre-Centre) of 1500mm generation a s/d ratio of 3.75 which meant the piles could be classed as a group. Additional consideration for the raft included thickness and overhangs so that the self-weight of the raft could be calculated and added on to the imposed load from the concrete arch.

Standard practice is to design an overhang of 150mm past each of the foundation piles and a thickness of at least 500mm however the overhangs can be 300mm thick as they are not carrying the main structural loads. Using these values and taking the density of concrete and appropriate steel reinforcement as 2500kN/m$^3$ the self-weight was calculated to be **28kN**.

Pile foundations' bearing capacity is generated through the base resistance of the pile & the sleeve friction. The comparative magnitude of these resistances is a function of diameter, depth, and present soil type. Firstly, we calculated base resistance which is defined by *Equation 7*.

$$Q_b = A_b C_{cpt} \overline{q_c} \qquad \text{(Eqn 7)}$$

As we had decided on an initial diameter (400mm) we first calculated the area of the base using the commonly known *Equation 8* before deriving the soil coefficient $C_{cpt}$ from *Table 1*.

$$A_b = \frac{\pi}{4} d^2 \qquad \text{(Eqn 8)}$$

*Table 1, (Knappett 2012)*

| **Pile Type** | **Soil** | $\boldsymbol{C_{cpt}}$ |
|---|---|---|
| Driven Piles (Closed Tip) | Sand | 0.4 |
| | Clay (Undrained) | 0.8 |
| | Clay (Drained) | 1.3 |
| Bored Piles | Sand | 0.2 |

The area of the base of each pile was calculated to be 0.126m$^2$, the soil coefficient was determined to be 0.4 as the base of each pile sits on top of a sandy layer of soil. This sandy layer was chosen as it has significantly higher bearing capacity than the clayey layers nearby. The next sandy layer was significantly deeper which would not have been efficient to construct.

Resultingly, we decided that the depth of the sandy layer should be the determining factor in the depth of our piles. They are to be constructed to a depth of 4000mm. Each pile should be fabricated off site to the desired specification to minimise wastage before being transported and driven into soil to the desired depth.

We recommend driving prefab concrete piles rather than boring to the required depth and then backfilling with concrete and reinforcement for two primary reasons:

- Bored concrete piles have a soil coefficient of 0.2, 50% of a comparative driven pile in sand. This means that their base resistance is also only 50% of their driven counterparts which means they would likely require a greater diameter which means more concrete, more reinforcement, more carbon & more costs.
- Construction of bored concrete piles is also likely to take significantly longer on site than driven piles. This is primarily because of the time to install reinforcement and pour concrete which then needs to be left for up to 28 days to reach its desired strength which is not necessary with prefab elements. There is also a higher chance of defects in the bored piles possibly requiring a higher factor of safety. More trades are also required.

The final component of the *Equation 7*, $\bar{q}_c$, refers to the average cone penetration resistance near the pile tip. This taken over a depth range equal to 1.5x the diameter of the pile, in our case **1200mm**. $\bar{q}_c$ was determined using *Equation 9* which can be thought of as an application of the statistical mean formulae.

$$\overline{q_c} = \frac{[\sum_{2.8 \leq d \leq 5.2} q_c]}{Number\ of\ q_c values\ in\ range} \qquad \text{(Eqn 9)}$$

The average cone penetration resistance was found to be 6.88 MPa. Inserting this alongside the other calculated values into *Equation 7* yields a base resistance, $Q_b$ of **347kN**.

As previously mentioned, the other component that generates pile capacity is the shaft resistance, this was determined by combining the surface area over the entire depth of the pile with $\mathrm{q_s}$ for each individual soil layer determined from *Tables 2&3*.

It is worth noting at this point that perhaps unintuitively there is no clear proven correlation between the shaft friction of the CPT probe and the shaft resistance of the pile. Instead, *Table 2* showing the '*Laboratiore des points et chaussees method*' has been used to determine an estimation of shaft resistance from the tip resistance of the CPT.

*Table 2, (Tomlinson 2001)*

| Soil Type | $q_s$ (Concrete) | $q_s$ (Steel) | Limiting $q_s$ (kPa) |
|---|---|---|---|
| **Silt & loose sand** | $\frac{q_c}{60}$ | $\frac{q_c}{120}$ | 35 |
| **Medium-dense sand & gravel** | $\frac{q_c}{100}$ | $\frac{q_c}{200}$ | 80 |
| **Dense to very dense sand & gravel** | $\frac{q_c}{150}$ | $\frac{q_c}{200}$ | 120 |

*Table 3*

| Soil Type | $q_c$ (MPa) | β for Driven/Jacked Concrete Piles | β for Driven/Jacked Metal Piles | Limiting $q_s$ |
|---|---|---|---|---|
| **Soft Clay** | <1 | 30 | 30 | 15 |
| **Firm-Stiff Clay** | 1-5 | 40 | 80 | 35 |
| **Stiff-Hard Clay** | >5 | 60 | 120 | 35 |

We then proceeded to apply the above tables to each layer that our pile foundation passed through:

- **Layer 1:** (0-1.7m), As $q_c$ is less than 1MPa according to *Table 3* it can be defined as soft clay and $q_s$ was taken as **15 kPa.**
- **Layer 2:** (1.7-2.4m), The average $q_c$ was 2.4Mpa and was classified as a sandy mixture using the soil profiles that we generated using the provided CPT data. Using *Table 2* and approximating the layer as a medium-dense sand we determined $q_s$ as **24kPa.**
- **Layer 3:** (2.4-4m), The average $q_c$ was 10.5Mpa and was classified as a sand using the same method as layer two. Using *Table 2* and approximating the layer as a medium-dense sand $q_s$ was determined as **104.9 kPa.**

After working out the resistance as a function of area to calculate the total shaft resistance each layer would have to be multiplied by its respective depth and then summated as shown in *Equation 10.*

$$q_{s(Total)} = \sum q_{s(layer)} \pi 2 r D_{(Layer)} \qquad \textit{(Eqn 10)}$$

This yielded a total shaft resistance of **263kN**. This could then be summated with the base resistance to give a total shaft capacity for each pile of **610kN.**

At this stage it is appropriate to check the capacity of foundation against the factored imposed load (FIL).

- The FIL, including both the arch loading and the concrete raft, is equal to **1528kN.**
- The foundation capacity is equal to the total capacity of each of the three piles which is **1830kN.**

Originally, we thought that we should re-iterate the design with a smaller pile diameter as the foundation system was only operating at 83% efficiency. However, there are also maximum settlement requirements imposed upon us by the client. This is relevant because as you approach the maximum capacity of a piled foundation the settlement increases at an exponential rate towards failure. If we reduced the diameter of the piles, they would operate more efficiently however the settlement increase would be very substantial and likely over the imposed limits.

A final capacity safety check is required for pile group foundation systems, this is known as checking against block failure. Block failure occurs when the entire section of soil below the raft fails as one giant 'pile'. Using *Equation 7* the base resistance for block failure was assessed and determined to be **2.7MN**. This is already greater than the total capacity of the piled foundation system so it is safe to determine that it will not fail in this manner and there is not a further requirement to check the block failure 'shaft resistance'.

# *Maximum Load: Settlement*

Settlement for the pile foundations has been calculated using the Hyperbolic Method, which is a function of the Pile Load, Pile Capacity, and the Young's Modulus of the soil at the base of the Pile.

This method uses several parameters:

- Pile Settlement, s
- Pile Load, Q
- Shaft Resistance, $Q_s$
- Base resistance, $Q_b$
- Soil Stiffness at base of Pile, $E_b$
- Pile shaft diameter, $D_s$
- Pile diameter at pile base, $D_b$
- Empirical Constant, $M_s$

The constant $M_s$ has been assumed to be 0.001.

These values have been used to find the following constants:

$$\alpha = Q_s \qquad \beta = D_b Q_b E_b \qquad \delta = 0.6 Q_b \qquad \lambda = M_s D_s \qquad \lambda = M_s D_s \qquad \textit{(Eqns 11 – 15)}$$

The parameters are subsequently used to find the coefficients a, b, and c to be used in a quadratic equation:

$$a = \eta(Q - \alpha) - \beta \qquad b = Q(\delta - \alpha) - \beta \qquad c = \lambda \delta Q \qquad \textit{(Eqns 16-18)}$$

Where:

$$as^2 + bs + c = 0 \qquad \textit{(Eqn 19)}$$

Since this is primarily a function of the Young's Modulus of the soil once all the capacities have been calculated, the next step included choosing an appropriate value for each soil type at the base of our piles.

The Young's Modulus values for the soils bearing the Pile Foundation and the Group pile have been estimated as 90Mpa for the sand layer at the base of the pile group. This assumption was based on typical values for their soil types and design charts made for the absence of sufficient data.

Only the positive values from the quadratic equations calculated were taken. These assumptions gave a settlement estimation of **15.04mm** for each pile in the pile group. This is well within the required limit of **30mm.**

# *Minimum Load: Foundation Design*

***Design Justification***

Contrary to our design for the maximum foundation load, for the minimum load we have opted for a single Pile Foundation. It would be unnecessary to design a pile group for a load that could be easily carried by one singular pile.

A preliminary depth of 5m was chosen, and this solution again avoids the need for very deep piles and takes advantage of a sand layer ~5m below the surface that provides significant bearing capacity.

***Capacity Calculation Methodology***

The lowest load acquired from WH Consultant was given as **90.46kN**, being resisted by a 0.3x0.3m concrete column. According to the CPT data given, this column is closest to CPT000012585, therefore all assumptions about the soil type has been taken from this data range. Consistent with the rest of the design, a safety factor of 2.5 has been used to account for uncertainties of the soil profile, leading to a Design load of **226.2kN.**

As previously mentioned, the Robinson Chart was used to indicate the soil layers present surrounding our foundation. Initial sizing of the pile foundation was assumed to be 0.5mx0.5m square driven concrete piles, at a depth of 5 metres. Due to the abundance of CPT data across the site, empirical methods were initially used to calculate the base and shear resistance of the pile.

Firstly, we calculated the Base Resistance as defined in *Equation 7.* For this method, the average cone resistance across 1.5 diameters from the base is taken, calculated to be **960kPa** from our CPT data. This pile primarily lies in a mixture of sandy layers, with its base resting just above a layer of silty SAND. $C_{cpt}$ was estimated using Table 1, (Knappett, 2012); due to a lack of data for a Soil Constant for silty sand, it was suggested that we could use a value between 0.4 for sand and 0.8 for clays. This was not the case in our final calculations, as it was decided to conservatively use the lower bound of 0.4.

Subsequently, the shaft resistance, $q_s$, of the pile has been calculated using *Tables 2 and 3* as previously shown, utilizing the equation for driven concrete piles in silt and loose sand.

$$q_s = \frac{q_c}{60} \qquad \textit{(Eqn 20)}$$

- **Layer 1:** (0-1.5m), in a sandy layer, $q_s$ can be calculated using Equation 20, giving a value of **9.3kPa**
- **Layer 2:** (1.5-4m), As $q_c$ is less than 1MPa according to *Table 3* it can be defined as soft clay and $q_s$ was taken as **15 kPa.**
- **Layer 3:** (4-5), The average $q_c$ was **2.15Mpa** and was classified as a sand using the same method as layer one. Using *Table 2* and approximating the layer as silt and loose sand $q_s$ was determined as **35kPa – the limiting value.**

By splitting the strata into layers and calculating the resistance across each layer of the soil, this yielded a total shaft resistance of **173kN**. This could then be summated with the base resistance to give a total resistance for the pile of **269kN.**

At this stage it is appropriate to check the capacity of foundation against the factored imposed load (FIL).

- The FIL, including both the arch loading and the concrete raft, is equal to **226.2kN**
- The foundation capacity is equal to the resistance of the pile, equal to **269kN.**

***Single Pile Design using Triaxial Data***

Alternatively, since this pile is primarily situated in a sandy layer, the data from several triaxial tests can be used in order to estimate the pile capacity.

The estimated coefficients from this data used with the initial assumptions for our pile diameter gave a pile capacity much larger than the design load, therefore for this set of data, we have reduced the size of the pile to a 400mm diameter circular pile to reduce the materials used but remains an appropriate size for column dimensions. The depth of this pile remains at 5m.

This method utilizes the internal angle of friction of the sand layers to estimate the base and shaft resistance for different types of piles. In a drained layer, the bearing capacity of the pile is a function of the *Cohesion Effect, Surcharge effect* and the *Soil self-weight* effect respectively.

$$q_b = s_c c'^{N_c} + s_q q_0 N_q + \frac{1}{2} s_\gamma \gamma' B N_y \qquad (Eqn21)$$

Due to the nature of the upper layers of soil, we can consider drained behaviour across the depth of the pile, therefore ignoring the cohesion effect, which is negligible in course soils. Furthermore, due to the small width of pile foundations, the self-weight effect of the soil can be effectively ignored for ease of calculation. Ignoring these values will of course only make the calculations more conservative.

The surcharge effect at the base is simplified to:

$$q_b = \sigma_v' \quad (Eqn\ 22)$$

The bearing capacity factor, $N_q$ can be estimated using the internal angle of friction from the Triaxial data, as well as the Berezantsev Chart for sandy soils (see below). With an internal angle of friction estimated to be **32°**, the bearing capacity factor is therefore approximately 43. Using the effective stress and the area of the pile, this gives a base capacity of **368.7kN.**

Calculating the shaft resistance of the pile uses a different method, as the shear strength of the soil is dependent on both the horizontal and vertical effective stresses present in the soil. The horizontal stress is found using the coefficient of lateral earth pressure and the vertical effective stress in drained soils. The coefficient of lateral earth pressure, $K_s$ is the interface friction angle due to the soil/pile interface, rather than the internal angle of friction: a soil/soil interface.

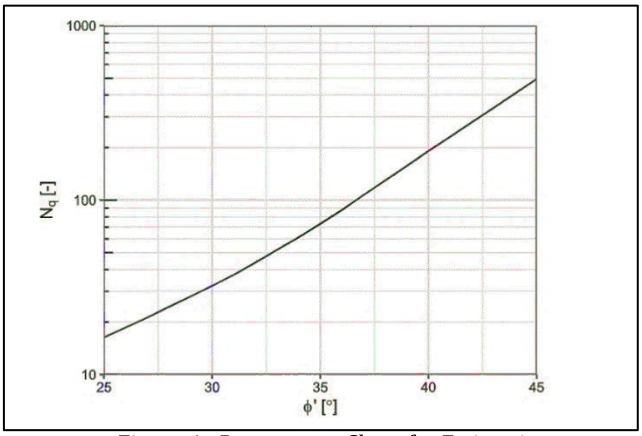



*Figure 6 - Berezantsev Chart for Estimating Bearing Capacity Factor, (Berezantsev, V. G, 1961)*

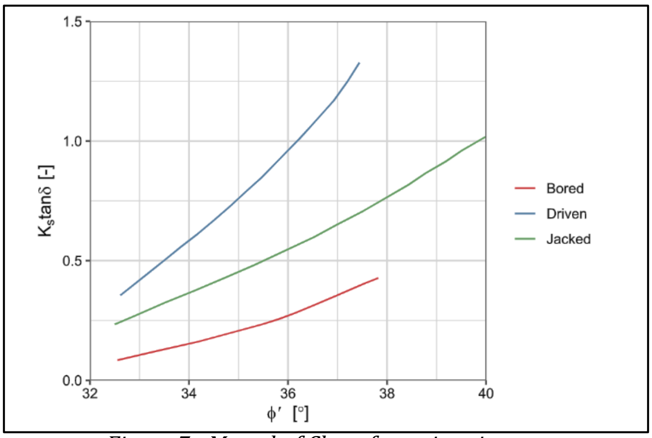



*Figure 7 - Meyerhof Chart for estimating Coefficient of Lateral Earth Pressure, (Meyerhof, G.G, 1976)*

It is estimated using the Meyerhof chart in *Figure 7*, which uses the friction angle to estimate both the coefficient of lateral earth pressure and angle delta as one value, found to be 0.4.

$$q_s = K_s tan\delta + \sigma_v' \qquad (Eqn\ 23)$$

Using the average vertical effective stress across the depth of the pile and the shaft area, the shaft resistance was calculated to be **92.2kN.**

This gave a massive final pile resistance of **478.9kN**, which is *far higher* than the pile capacity needed, therefore the decision was made to use the **Pile Capacity calculated using the CPT data,** which is to say, a base resistance of **269kN**. Since the triaxial test was completed using one type of soil, it is perhaps not representative of the wide range of soil types present, and it was more feasible to use the abundant range of CPT data to take a more conservative approach towards our pile capacity, and therefore its settlement.

# *Minimum Load: Settlement*

The settlement for the minimum load was calculated the exact same method as the pile group: using the hyperbolic method. This involved using the dimensions of 0.5x0.5m of the pile, the Design Load, and the Base and Shaft Resistance. The pile has been shown to rest on a silty SAND layer, therefore the Young's modulus at the base has been estimated to be **15MPa**. This conservative estimate was undertaken due to the mix of fine grained and course soils, making it an important decision when using the appropriate design chart.

Using these parameters, a final settlement of the pile was found to be **12.69mm,** which is once again well within the limit of 30mm, and very close to the settlement calculated for our pile group. The small difference between settlement across the site will reduce the possibility of adverse stresses caused by differential settlement.

# *Overall Conclusions & Recommendations*

The foundation design was heavily influenced by the CPT and triaxial data analysis, rather than the oedometer data, which was deemed unrepresentative of the strata as a whole.

For the maximum loading of **1528kN**, we decided to design a pile group with base areas of **0.126m$^2$** each. We chose to design the pile group to a depth of **4000mm** so that each base could sit on a sandy layer to take advantage of its higher bearing capacity. We also decided it would be beneficial to drive the piles rather than boring them to ensure a larger base resistance, smaller diameter, reduced concrete, faster construction time and lowered chance of defects. Our designed pile group passed its capacity safety check with a total foundation capacity of **1830kN** and therefore can withstand the maximum loading of our building.

A pile group was deemed unnecessary for our minimum load of **226.2kN** and therefore we opted for **500mmx500mm** square driven single pile foundations with a depth of **5000mm**. We decided to take a more conservative approach and use the total pile resistance calculated using the CPT data (**269kN**) rather than the triaxial data (**478.9kN**) for our single pile foundation's capacity check. Both foundation designs for the maximum and minimum loads of our building sit comfortably within settlement limits (30mm) at **15.04mm** and **12.69mm**.

# ***References***

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